

MERVEILLES AGBETI-MESSAN

PhD Student in Artificial Intelligence, LITIS, Université de Rouen Normandie

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RESEARCH INTERESTS

Document Intelligence and Layout Analysis | OCR/HTR for Historical Documents | Efficient Sequence Models (State-Space Models / Mamba) | Multimodal Document Understanding | Knowledge Graphs and LLMs for Information Extraction

EDUCATION

PhD in Artificial Intelligence, LITIS Laboratory Nov. 2024 – Present

Université de Rouen Normandie *Rouen, France*

- Research on OCR and structure understanding for historical newspapers; three papers published at ICDAR 2026, including one Best Paper Award. Advisors: Prof. Thierry Paquet, Prof. Clément Chatelain.

M.Sc. Operational Research and Optimization Sep. 2023 – Oct. 2024

École Nationale de l'Aviation Civile (ENAC) *Toulouse, France*

- Mathematical optimization for machine learning, deterministic global optimization, advanced combinatorial optimization.

M.Eng. Computer Science (Distributed Systems and Big Data) Sep. 2021 – Oct. 2024

Institut National des Sciences Appliquées (INSA) *Toulouse, France*

- GPA 3.84/4.0, ranked top 5%. Machine learning, deep learning, big data infrastructures, cloud computing.

Classes Préparatoires, Mathematics and Physics Sep. 2019 – Aug. 2021

CPGE Ibn Ghazi *Rabat, Morocco*

- Excellence Scholarship; ranked 1st in Computer Science.

PUBLICATIONS

Author name in bold. All papers at ICDAR 2026, the flagship venue for Document Analysis and Recognition.

A Benchmark of State-Space Models vs. Transformers and BiLSTM-based Models for Historical Newspaper OCR

M. Agbeti-Messan, P. Tranouez, S. Nicolas, C. Chatelain, T. Paquet.

ICDAR 2026 | **Best Paper Award** | First author | [arXiv:2604.00725](https://arxiv.org/abs/2604.00725) | [code](#)

Scaling State-Space Models from Lines to Paragraphs: An Ablation of Mamba-based OCR

M. Agbeti-Messan, P. Tranouez, S. Nicolas, C. Chatelain, T. Paquet.

ICDAR 2026 (WML workshop) | First author | [arXiv:2606.23524](https://arxiv.org/abs/2606.23524)

Towards Hierarchical Structure Understanding of Newspaper Images

W. Mocaër, S. Tarride, T. Constum, **M. Agbeti-Messan**, T. Simon, C. Chatelain, S. Nicolas, P. Tranouez, S. Cretin, T. Paquet.

ICDAR 2026 (HIP workshop) | [arXiv:2607.15082](https://arxiv.org/abs/2607.15082)

RESEARCH EXPERIENCE

Doctoral Researcher Nov. 2024 – Present

LITIS, Université de Rouen Normandie *Rouen, France*

- Benchmarked State-Space Models (Mamba) against Transformer and BiLSTM baselines for end-to-end OCR of historical newspapers, showing that linear-time SSMs match Transformer accuracy on printed historical text; the study won the Best Paper Award at ICDAR 2026.
- Designed and ran a systematic ablation of Mamba-based OCR from line to paragraph level, isolating the recurrent state dimension and expansion factor as the dominant drivers of long-sequence accuracy; published as first author at ICDAR 2026 (WML).
- Contributed to methods for hierarchical structure understanding of newspaper images, contributing to a paper published at ICDAR 2026 (HIP).

- Released open benchmark code alongside the Best Paper to make the results reproducible for the community.

AI Research Engineer, Data Scientist (Internship)

Feb. 2024 – Aug. 2024

Capgemini Engineering

Blagnac, France

- Tackled the low domain accuracy of generic LLMs on aerospace maintenance texts by building a pipeline that combines LLMs and Knowledge Graphs, improving named-entity recognition and delivering a working research prototype.
- Surveyed the state of the art and engineered data-preprocessing steps to raise data quality, then documented methods and findings for scientific reporting.

ENGINEERING EXPERIENCE

Software Engineer (Internship)

Jun. 2023 – Sep. 2023

Sopra Steria

Colomiers, France

- Built Java services and SQL scripts for the French Ministry of Agriculture (PAGODE, LOGICS) to streamline digital declarations, and authored 100+ Jira test cases that secured the reliability of fishing-permit requests.

Software Engineer (Internship)

Jun. 2022 – Sep. 2022

WeBuildUp

Versailles, France

- Refactored 20+ Next.js/TypeScript components to lift code readability by 10%, wrote 80+ unit, integration and end-to-end tests (Jest, Playwright), and delivered the company website.

Python Developer (Internship)

Aug. 2019 – Sep. 2019

StaroCity

Lomé, Togo

- Built a PyQt5 desktop application for Wi-Fi credential management that eliminated duplicate accounts and sped up account creation using Python decorators.

OPEN-SOURCE AND PROJECTS

react-microsoft-clarity | *TypeScript, React, npm*

- Authored and maintain an open-source React component for Microsoft Clarity with roughly 35,000 weekly downloads on npm.

ssm-ocr-benchmark | *PyTorch, OCR, State-Space Models*

- Released the benchmark code accompanying the ICDAR 2026 Best Paper for reproducible comparison of SSM, Transformer and BiLSTM OCR models.

Multi-class Image Classification | *CNN, ResNet50, InceptionV3*

- Trained transfer-learning models for image classification and wrapped them in a full front-end application.

HONORS AND AWARDS

Best Paper Award, ICDAR 2026 | **2nd place**, ICPC international collegiate programming contest | **Best Presentation Award**, Junior INSAkathon 2023 (Sports and Disability) | **Excellence Scholarship**, Government of Togo | Ranked **1st** at the national Computer Science entrance competition (Togo) | Baccalauréat with Highest Honours

TECHNICAL SKILLS

Languages: Python, Java, C, OCaml, JavaScript/TypeScript, SQL, R

ML / Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas; Transformers, State-Space Models (Mamba), CNNs, LLMs

Document AI: OCR/HTR, PAGE XML, CVAT, Layout Analysis

Data and Cloud: Neo4j, PostgreSQL, MongoDB, Hadoop, Spark, Docker, OpenStack, Azure

Web: React, Next.js, Node.js, Angular, TailwindCSS

Spoken Languages: French (native), Éwé (native), English (professional), Japanese (basic)